TwinSense AI — System Architecture Specification

Predictive Healthcare Digital Twin Prototype | 100% Synthetic Multi-Patient Simulation | Local-First Zero-Cloud Architecture

FastAPI 0.110+ | Scikit-Learn 1.4+ | Argon2id Auth | ReportLab PDF | 123 Pytest Passing | MIT License

PHASE 1: OFFLINE DATA & TRAINING PIPELINE (BATCH)

1. Synthetic Patient Data Generator

Seeded: 42

src/data_generator.py (Deterministic Simulation)

- Models 10 clinically diverse patients (T1D & T2D)
- 7 days continuous CGM @ 5-min intervals (20,160 readings)
- Physiological range enforcement: [40.0, 400.0] mg/dL

Output Raw Datasets:

```
data/raw/patient_profiles.csv (10 profiles)
data/raw/glucose_timeseries.csv (20,160 rows)
```

2. Preprocessing & Feature Engineering

11 Features

src/feature_engineering.py (Past-Only Windows)

- Joins static EHR baselines (Age, BMI, HbA1c, Duration)
- Extracts past-only rolling features (15m/30m means, RoC)
- Activity telemetry: 30m step sum, 30m mean heart rate
- Target formulation: Glucose state at t + 45 minutes

Temporal Leakage Prevention Protocol:

- Chronological 80/20 partition per patient
- 45-minute embargo buffer dropped at partition boundary

```
data/processed/training_dataset.csv (20,020 rows)
```

3. Random Forest Model Training

89.29% Acc

src/train.py (Chronological Split + Embargo)

- Model: Balanced RandomForestClassifier (100 trees, depth 8)
- Benchmark: DummyClassifier (most frequent baseline)
- Macro F1 on active classes: 0.8917 (vs Baseline: 0.3557)
- Accuracy: 89.29% on untouched holdout test set (N=3,920)

Honest Limitation Guardrail:

- Class 1 (Hypoglycemia) support: 0 samples in train & test
- Preserves zero-reassurance constraint & warning policy

4. Persisted Model & Schema Artifacts

Joblib / JSON

models/ & reports/ (Serialized Assets)

Persisted Trained Artifacts:

```
models/random_forest_risk_model.joblib
models/feature_columns.json (Strict 11 features)
```

Audited Documentation:

```
reports/evaluation_report.md (Metrics & matrices)
reports/model_evaluation.md (Comparative audit)
```

Trained Model Weights & Feature Schema Loaded on Startup

Idempotent Database Initialization (Raw CSVs → SQLite on Startup)

PHASE 2: ONLINE RUNTIME PREDICTION & INTERACTIVE TWIN (LOCAL SERVER)

9. Login & Registration System

Argon2id

app/auth.py & app/static/ (Security Layer)

- RFC 9106 Argon2id password hashing (salted)
- Never stores or returns plain-text passwords
- Server-managed secure sessions in SQLite
- HttpOnly twinsense_session cookie (SameSite)
- Double-submit twinsense_csrf token protection
- Non-revealing error messages against user enum

```
/login.html | /register.html | /api/auth/*
```

10. HTML/CSS/JavaScript Dashboard

Vanilla JS & SVG

app/static/ (Pure Native Web Client)

Self-Contained Frontend (Zero External CDNs):

- index.html: Semantic UI with glassmorphic cards
- style.css: Modern clinical violet/teal theme
- app.js: Pure ES6 controller, native SVG chart engine

Interactive Core Capabilities:

- Patient Selector: Switch profiles (PAT_001 - 010)
- High-Frequency CGM Chart: Real-time SVG time-series
- 45-Min Forward Forecast Panel: In-Range vs Hyper
- What-If Counterfactual Sandbox: Delta sliders
- Clinical Explanations: Key contributing factors
- Real-time Low-Glucose Rule Alert (< 70 mg/dL)
- Persistent Disclaimers: Non-clinical synthetic data
- One-Click Clinical Report Download (CSV / PDF)

HTTP / JSON REST Calls

Typed API Responses

5. FastAPI Backend Application

Port 8000 / ASGI

app/main.py, schemas.py, services.py

Application Core & Lifespan Manager:

- Idempotent startup seeder from raw CSV data
- Static file mounting for dashboard & auth pages
- CORS middleware restricted to local origins

Typed REST API Routes (Pydantic v2):

```
GET  /api/health (System status & warnings)
GET  /api/patients (List 10 profiles)
GET  /api/patients/{id}/readings (Telemetry)
POST /api/patients/{id}/predict (45m Inference)
POST /api/patients/{id}/what-if (Simulation)
GET  /api/patients/{id}/predictions (Audit log)
GET  /api/patients/{id}/report/csv (Export)
GET  /api/patients/{id}/report/pdf (Export)
```

Inference & What-If

Risk Probs & Factors

6. SQLite Local Relational Store

SQLite3

data/twinsense.db (Zero-Cloud Local DB)

Relational Schema Tables:

- patients: 10 synthetic baseline EHR records
- telemetry_readings: 20,160 5-min sensor records
- users: Registered clinician/researcher credentials
- sessions: Active server-side sessions with expiry
- predictions: Append-only forecast audit log

Operational Guarantees:

- Idempotent CSV population on cold start
- Indexed foreign keys (patient_id, timestamp)
- What-if simulation does NOT write to database
- Safe local storage without external network calls

7. TwinEngine Digital Twin Core

Singleton

src/twin_engine.py (In-Memory Engine)

In-Memory Digital Twin Representation:

- Couples static EHR profile with dynamic CGM stream
- Validates temporal ordering & min 30m sensor window
- Evaluates Random Forest model via scikit-learn

What-If Counterfactual Sandbox:

- Ephemeral in-memory telemetry clone (Zero DB write)
- Evaluates Delta Glucose (-50 to +50 mg/dL)
- Evaluates Delta Activity (0 to +1,500 steps)
- Generates side-by-side observed vs simulated risk

8. Safety Alerts & Suggestions

Safety Engine

app/suggestions.py & src/twin_engine.py

Rule-Based Low-Glucose Safety Alert:

- Trigger: Measured glucose < 70 mg/dL (Rule-based)
- Independent of ML; fires immediately without forecast
- Zero insulin/dosing advice; clear emergency guidance

Transparent Model Limitation Warnings:

- 'Hypoglycemia prediction is not supported by model'
- Class 1 displayed as 'Not supported' (never 0%)

Contextual Lifestyle & Clinician Review Suggestions

11. Prediction History & Reports

ReportLab PDF

app/reports.py (Clinical PDF / CSV Export)

Historical Audit Trail (/api/patients/{id}/predictions):

- Traceable forecast history with timestamps & factors

CSV Clinical Audit Report (/report/csv):

- Tabular dump of probabilities, features, and alerts

PDF Clinical Summary Report (/report/pdf):

- Professional ReportLab vector PDF generation
- Patient EHR header, risk breakdown, and alert box
- Prominent non-clinical synthetic data disclaimers

Export Request

Binary PDF / CSV Stream